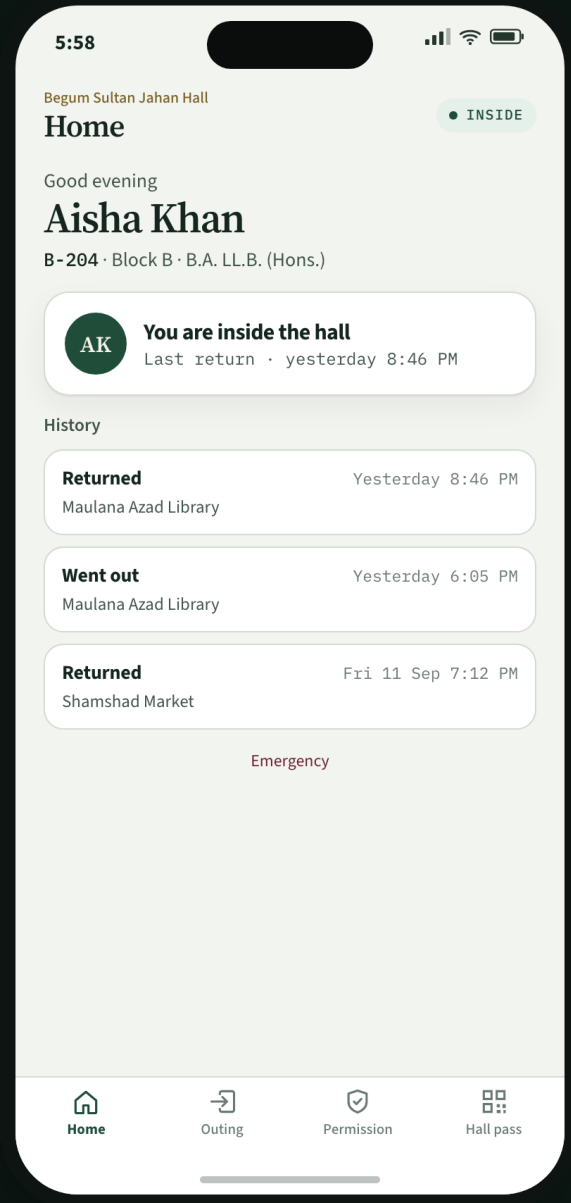
Begum Sultan Jahan Hall

Aligarh Muslim University

Hostel Management App Proposal

Built for

- Resident
- Warden
- Gate staff
- Provost



The case

Today

Movement is a paper register at the gate.

With the app

- Students and admins check in and check out from one app.
- Residents request special permission.
- Wardens approve special permission.
- Parents are notified if a resident is late.

How the hall pass works

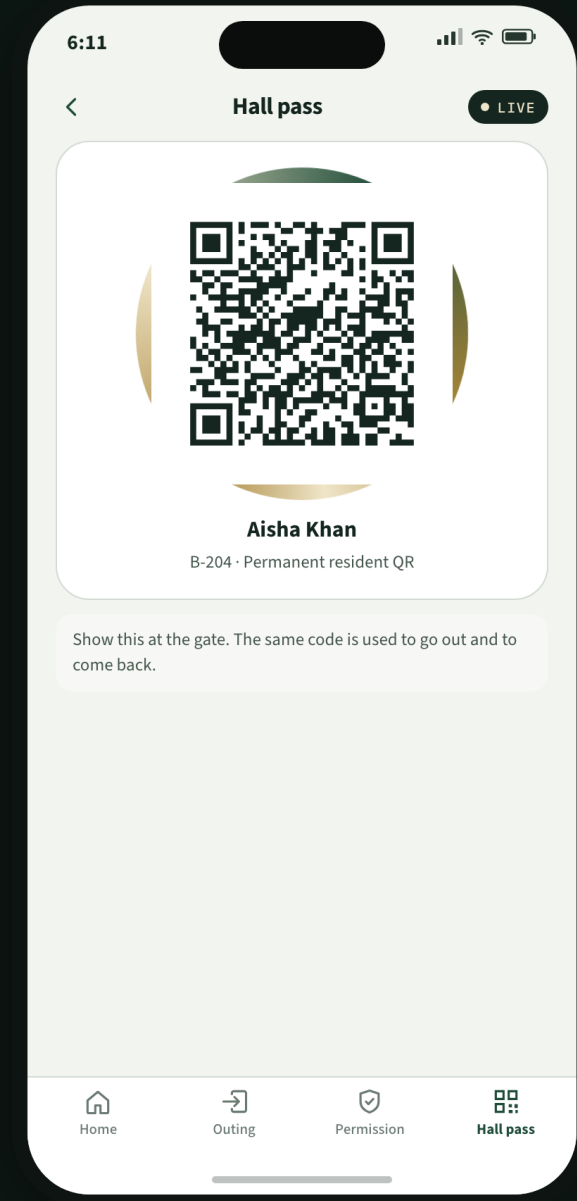
Every resident has a hall pass on the phone: a QR code. At the gate, staff scan this hall pass. The app already knows whether that resident is inside the hall or outside.

If the resident is inside the hall

The resident shows the hall pass. Gate staff scan it. The app records that the resident has gone out. Destination and return time were already set in the app.

If the resident is outside

The resident shows the same hall pass. Gate staff scan it. The app records that the resident has come back. Nothing extra to fill in.



Built for everyone

Resident

Plans an outing, requests special permission, and shows the hall pass at the gate.

Warden

Sees who is inside, outside, or late. Approves or rejects special permission.

Gate staff

Scans the resident hall pass. The app records exit or return.

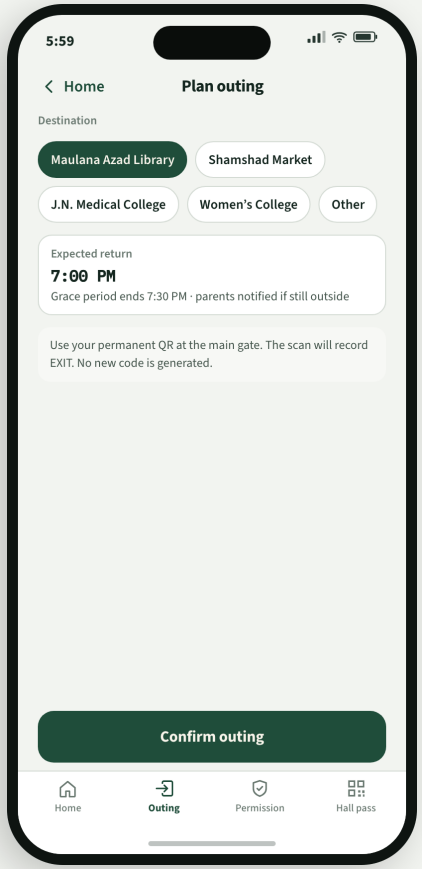
Provost

Sees the whole hall. Can export the register.

Scenario 1. A resident wants to go out

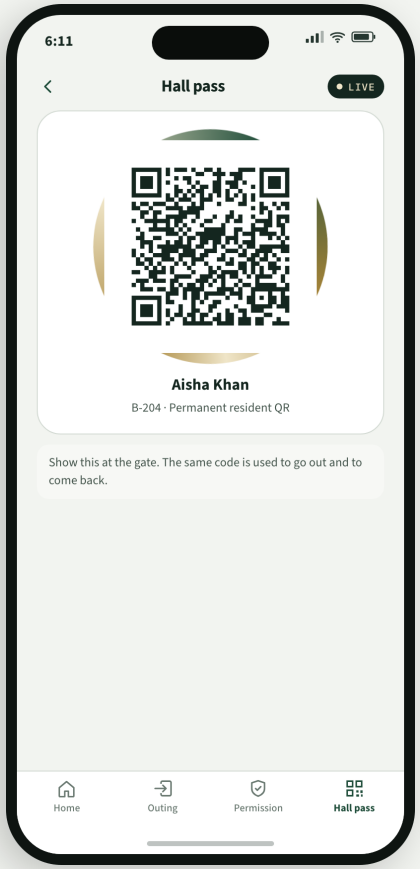
Resident

1. The resident sets destination and return time.



Resident

2. The resident opens the hall pass at the gate.



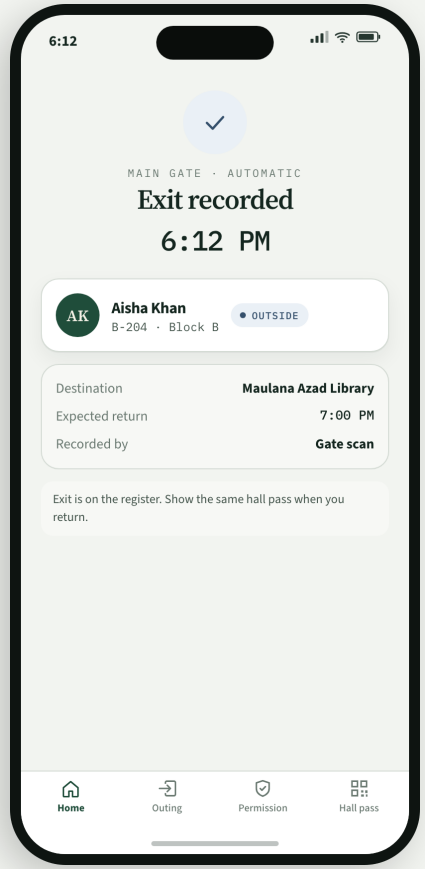
Gate staff

3. Gate staff scan the hall pass.



Resident

4. Exit is recorded. The resident is now outside.



Scenario 2. A resident comes back

Resident

1. The resident opens the same hall pass.

Gate staff

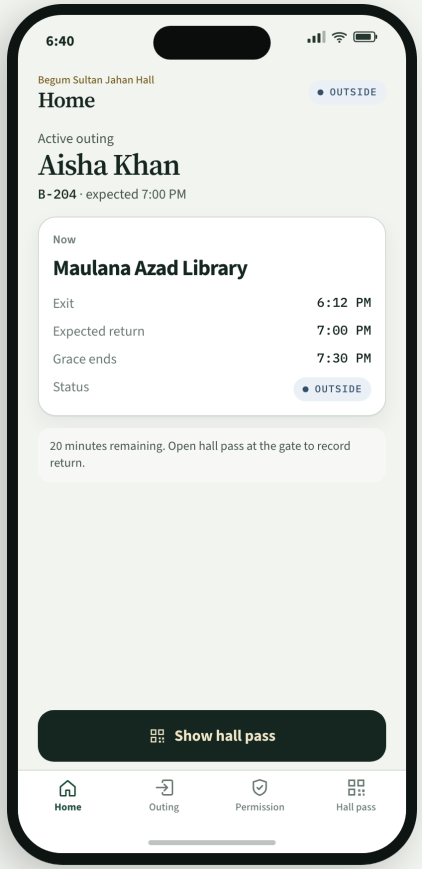
2. Gate staff scan it. Return is recorded.

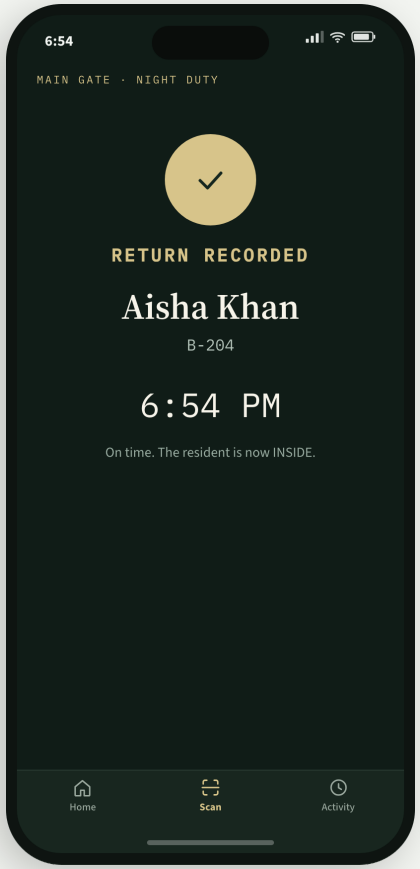
Warden

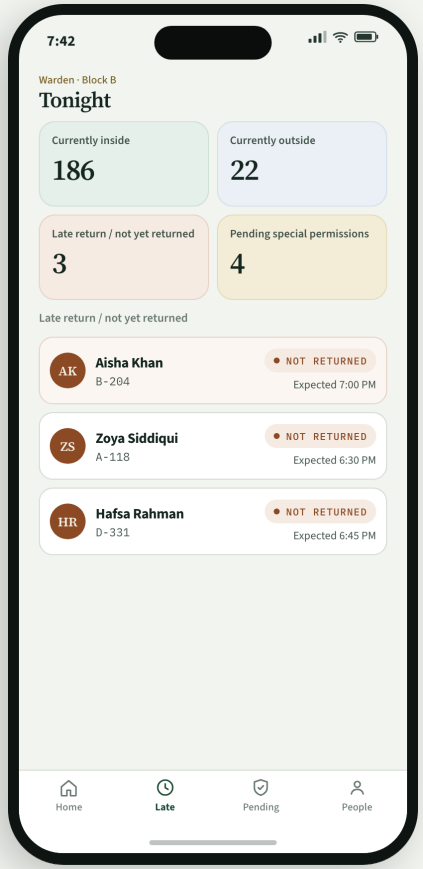
3. If the resident is late, the warden sees the late list.

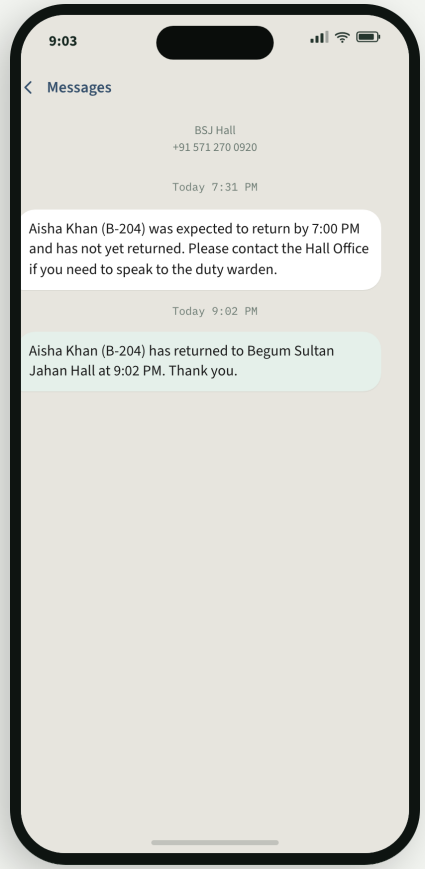
Parent phone

4. A parent SMS is sent, and another when the resident returns.





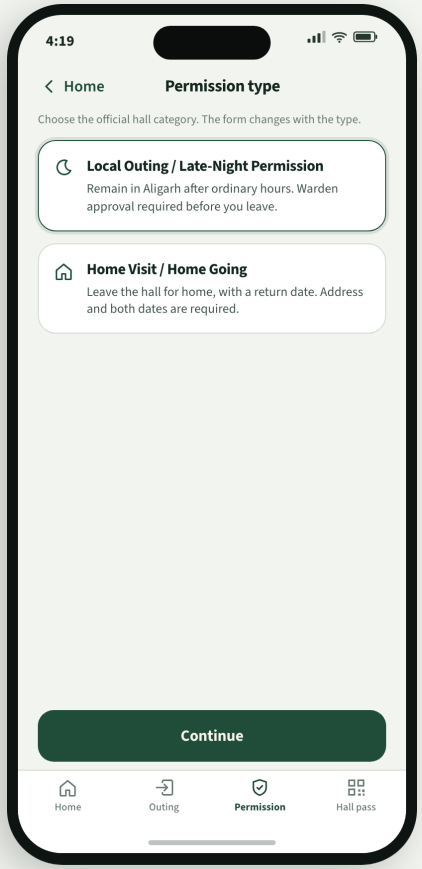




Scenario 3. Special permission

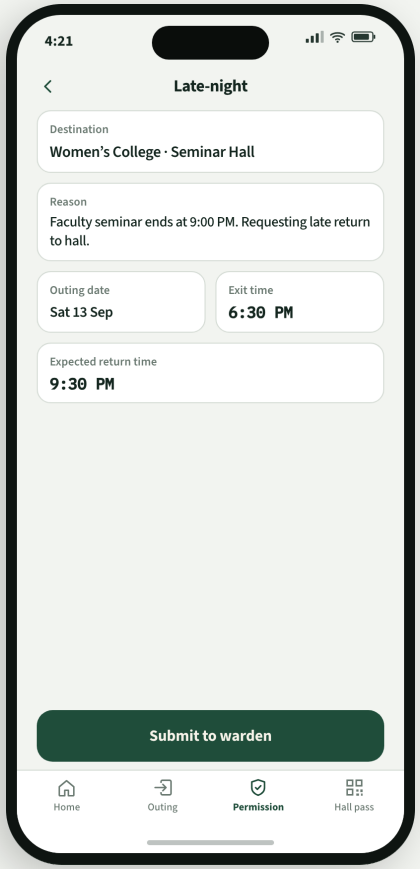
Resident

1. The resident chooses late-night or home going.



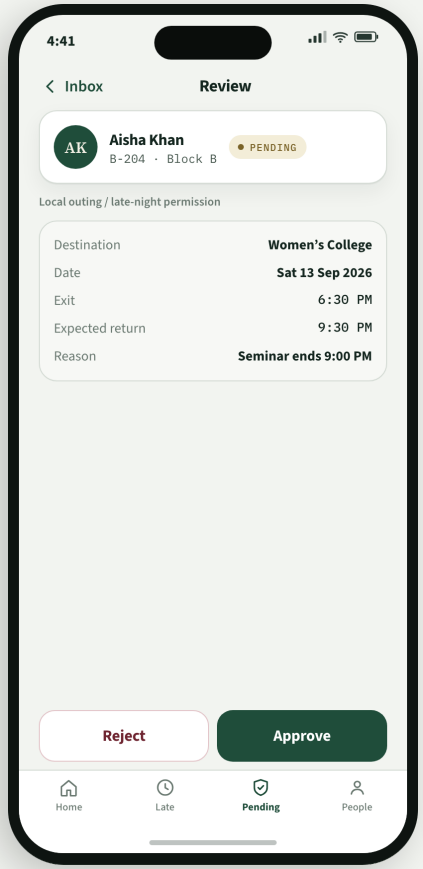
Resident

2. The resident fills the form and sends it to the warden.



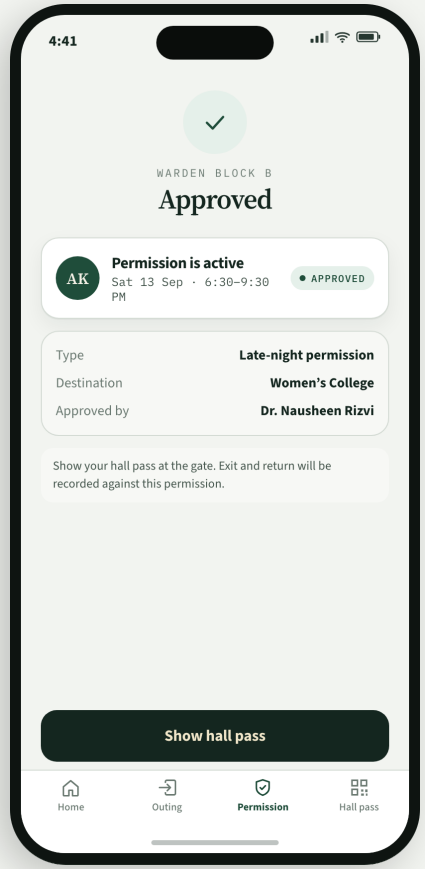
Warden

3. The warden approves or rejects.



Resident

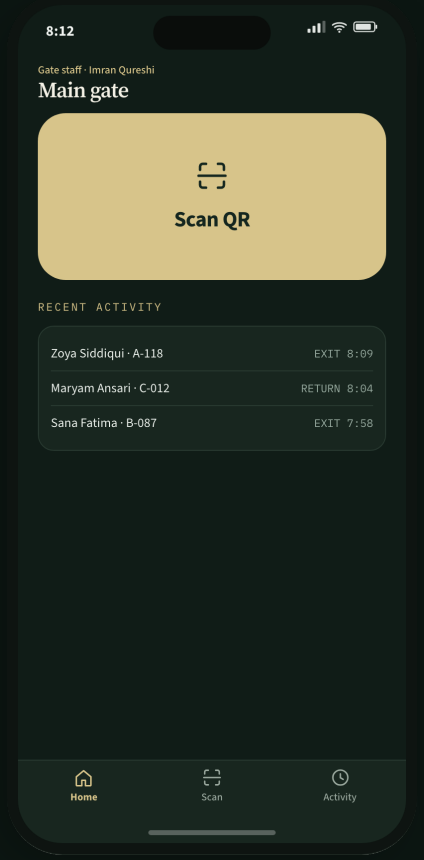
4. If approved, the resident uses the same hall pass at the gate.



Scenario 4. Gate staff

Gate staff

1 Gate staff open Scan QR.



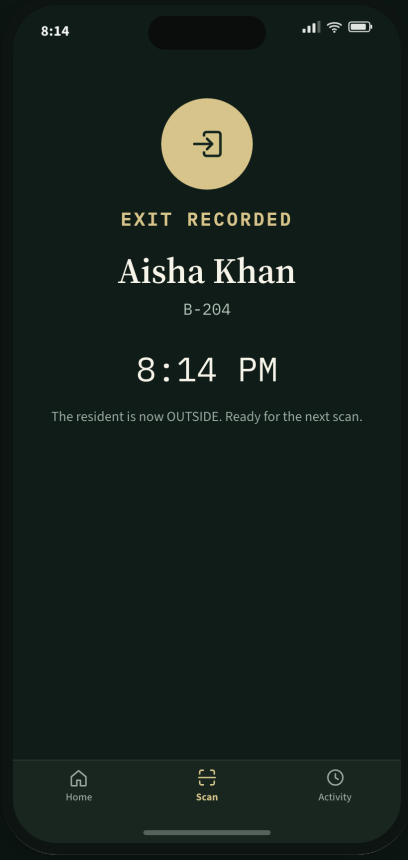
Gate staff

2 Gate staff scan the resident hall pass.



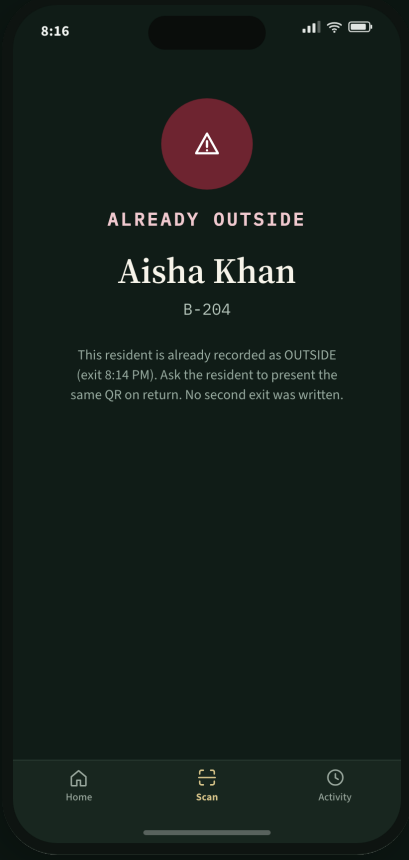
Gate staff

3 The screen shows Exit recorded or Return recorded.



Gate staff

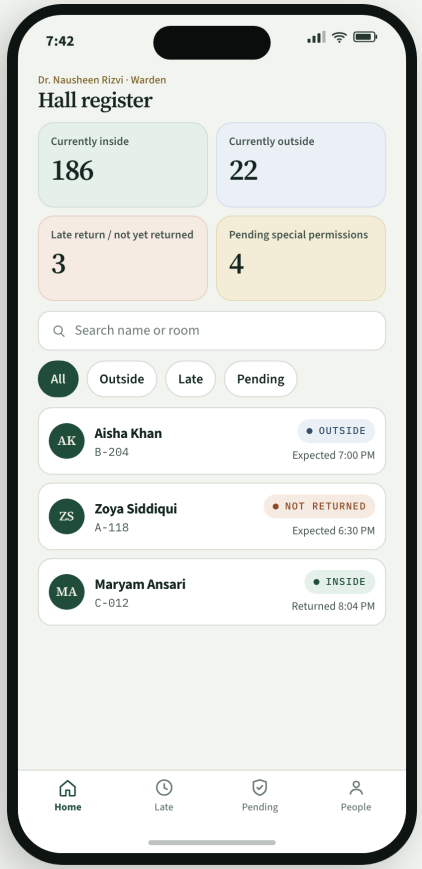
4 If the scan cannot be used, the screen says what to do.



Scenario 5. Warden

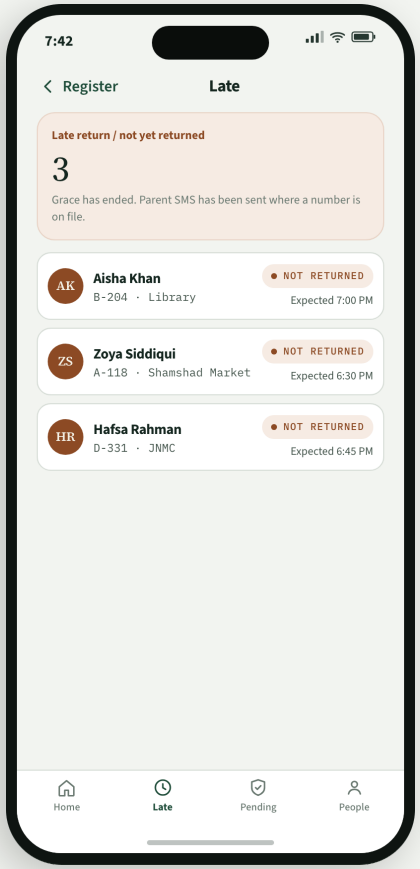
Warden

1 The warden opens the hall register.



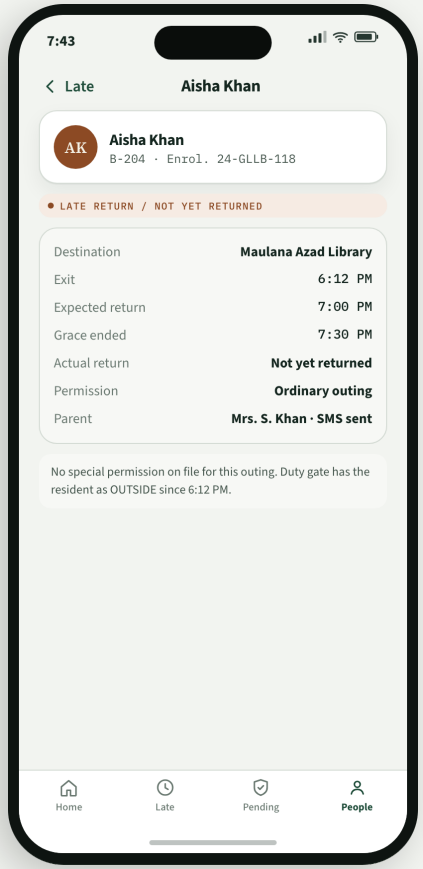
Warden

2 The warden opens the late list.



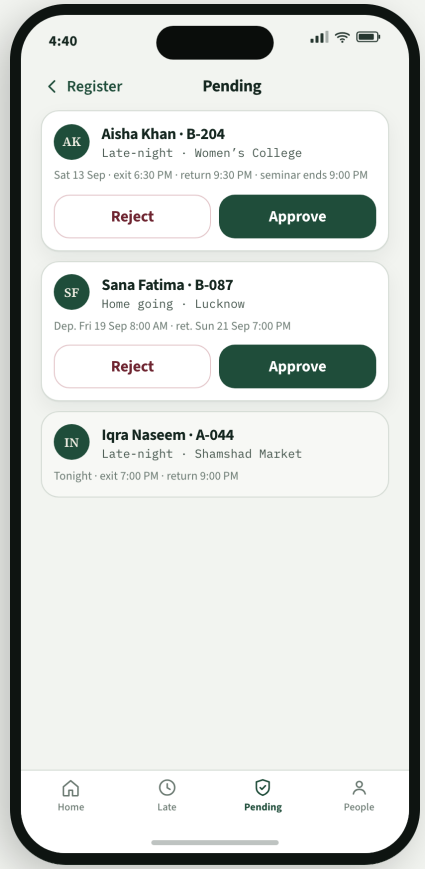
Warden

3 The warden opens a resident record.



Warden

4 The warden approves or rejects special permission.

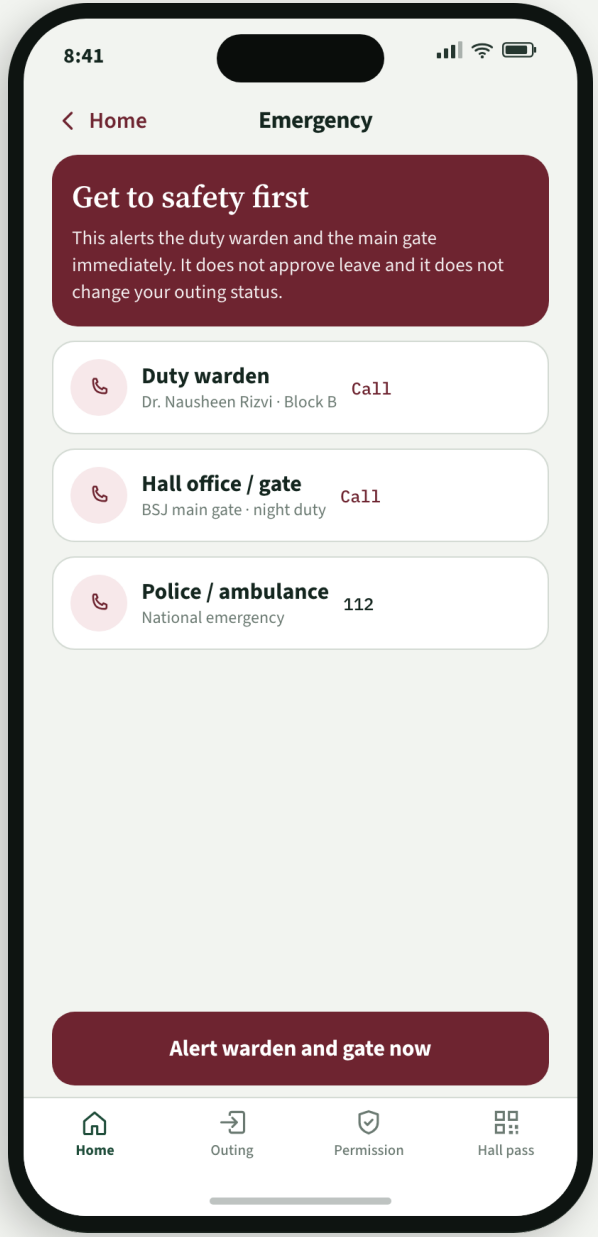


Optional

Emergency

A resident can alert the duty warden and the gate, and call from the phone. This is not a leave form.

Skip this if the hall does not want it in the first version.



What we will build

1. Ordinary outing: destination, return time, scan at the gate
2. Return, including late return and parent SMS
3. Special permission: late-night and home going
4. Gate scanning
5. Warden register: inside, outside, late, pending

Optional: emergency contacts on the resident phone. Not in this version: mess, rooms, fees, biometrics, a parent app.

Running cost

Each year, to keep the app live. Developer cost to be confirmed.

Item	Each year
India server	₹6,000
Parent SMS (only when a resident is late)	₹2,000
App Store (personal Apple Developer account)	₹0
Play Store (personal Play account)	₹0
Total each year	**₹8,000**

App Store and Play Store can use the existing personal accounts, same as the ICACON Companion app.

Optional fixes only when used: ₹2,000 per hour.